Replicating and Extending Card & Krueger (1994): Minimum Wages and Employment in the Fast-Food Industry

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Abstract

This report replicates the classic difference-in-differences study of Card and Krueger (1994), which estimated the employment effect of New Jersey's 1992 minimum-wage increase using fast-food restaurants in Pennsylvania as a control group. Our replication recovers the paper's headline estimate almost exactly—a difference-in-differences of 2.75 full-time-equivalent employees per store, against the paper's 2.76—and confirms the paper's surprising conclusion that the wage increase did not reduce employment. We extend the analysis with a first-stage wage check, heterogeneity by store ownership, alternative employment outcomes, and robustness to clustered standard errors and outliers.

1 Research question and motivation

Research question. Does raising the minimum wage reduce employment?

This is one of the longest-standing questions in labour economics. The textbook competitive model predicts that a binding minimum wage—one set above the market-clearing wage—moves firms up the labour-demand curve and *lowers* employment. Before 1994 this view dominated both theory and policy debate, yet the existing evidence was largely time-series and observational, and therefore could not cleanly separate the effect of the minimum wage from the many macroeconomic forces moving employment at the same time.

Motivation and identification. The authors exploit a natural experiment: on April 1, 1992, **New Jersey raised its state minimum wage from $4.25 to $5.05** per hour, while neighbouring **Pennsylvania kept it at $4.25** (the federal level). Fast-food restaurants on either side of the state border—Burger King, KFC, Roy Rogers, and Wendy's—are nearly identical in technology and labour demand, so Pennsylvania restaurants form a natural counterfactual for what New Jersey employment *would have* done in the absence of the hike. The result is one of the earliest and most influential difference-in-differences (DID) designs in economics.

The paper's finding—that the minimum-wage increase was **not** associated with employment loss, and may even have raised employment—was startling at the time and remains central to modern minimum-wage debates.

2 Data sources, sample construction, and variable definitions

2.1 Data source

The data are Card and Krueger's original survey of fast-food restaurants in New Jersey and eastern Pennsylvania, conducted in **two waves**: wave 1 in February–March 1992 (before the April 1 wage hike) and wave 2 in November–December 1992 (after). The file `data/public.csv` contains **410 stores** × **46 columns**. Columns with the suffix 2 are second-wave responses; blank cells are missing values.

2.2 Sample construction

Table 1: Sample construction

Step	Rule	Stores
Wave-1 file	all observations	410
Second-wave status	`STATUS2 = 1` (answered) or `= 3` (permanent closure)	405
Valid employment both waves	`FTE1`, `FTE2` non-missing	**384** (NJ 309, PA 75)

Permanently closed stores are kept at **zero employees**—the raw data already records them as $0, 0, 0$. Excluding them would introduce *survivorship bias* (stores that closed under wage pressure would vanish from the "after" average, inflating it). Stores temporarily closed, refused, or otherwise unobservable in wave 2 are dropped.

2.3 Variable definitions

Table 2: Variable definitions

Variable	Definition
`FTE`	Full-time-equivalent employment = full-time +0.5× part-time + managers
`DFTE`	Change in FTE (wave 2 − wave 1)
`NJ` (= `STATE`)	1 = New Jersey (treated), 0 = Pennsylvania (control)
`GAP`	Wage gap: (5.05−WAGE_ST)/WAGE_ST for NJ stores with starting wage below \$5.05; 0 for PA and compliant NJ stores; missing if the starting wage is unknown
`CHAIN`	1 = Burger King, 2 = KFC, 3 = Roy Rogers, 4 = Wendy's
`CO_OWNED`	1 = company-owned (vs. franchise)
`SOUTHJ`, `CENTRALJ`, `NORTHJ`, `PA1`, `PA2`, `SHORE`	Region indicators (South/Central/North NJ, two PA regions, NJ shore)

3 Empirical design

3.1 Difference-in-differences

The core specification regresses each store's change in employment on the treatment dummy, optionally with controls:

$$\mathrm{DFTE}_i = \alpha + \beta\,\mathrm{NJ}_i + \gamma' X_i + \varepsilon_i. \tag{1}$$

The coefficient β is the DID estimate. Equivalently, in means form:

$$\mathrm{DID} = \big(\mathbb{E}[\mathrm{FTE2} \mid \mathrm{NJ}] - \mathbb{E}[\mathrm{FTE1} \mid \mathrm{NJ}]\big) - \big(\mathbb{E}[\mathrm{FTE2} \mid \mathrm{PA}] - \mathbb{E}[\mathrm{FTE1} \mid \mathrm{PA}]\big). \tag{2}$$

Identifying assumption. Parallel trends: absent the wage hike, NJ and PA employment would have evolved identically. With two survey waves this cannot be tested directly; the design's credibility rests on the geographic and sectoral similarity of the two groups.

3.2 Treatment-intensity (wage-gap) reduced form

The paper's second design uses the *bite* of the law rather than a binary treatment. For NJ stores, `GAP` is the proportional increase needed to reach the new minimum. Regressing the employment change on `GAP`,

$$\mathrm{DFTE}_i = \alpha + \theta\,\mathrm{GAP}_i + \gamma' X_i + \varepsilon_i, \quad i \in \text{New Jersey}, \tag{3}$$

estimates the employment response to the *intensity* of the mandated wage increase.

3.3 Estimation details

All regressions use heteroskedasticity-robust (HC1) standard errors; a `matrix_rank` assertion guarantees the region controls are not collinear with the treatment dummy.

4 Core replication results and comparison with the original paper

4.1 Table 3: mean FTE employment per store

Table 3: Mean full-time-equivalent employment per store, before and after (standard errors in parentheses)

	n	Before	After	Change
New Jersey	309	20.43 (0.52)	20.90 (0.53)	+0.47 (0.48)
Pennsylvania	75	23.38 (1.39)	21.10 (0.97)	−2.28 (1.25)
Difference-in-differences				**+2.75**

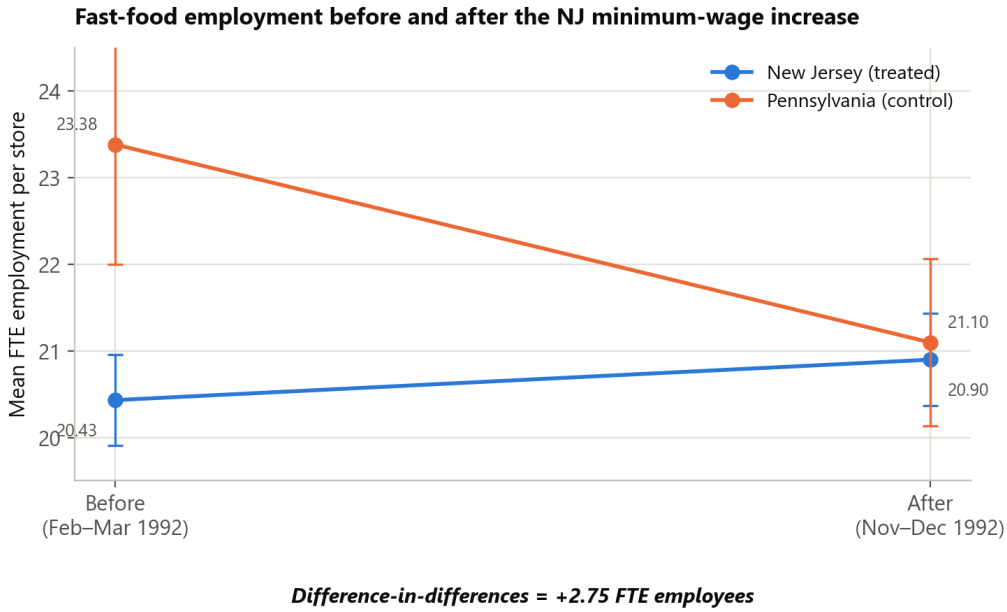


Figure 1: Fast-food employment before and after the New Jersey minimum-wage increase

Comparison with the paper. Card and Krueger report NJ before = 20.44, NJ change = +0.59, PA change = −2.16, and DID = +2.76 (SE 1.36). Our DID of **2.75** matches to the

second decimal, and the NJ pre-treatment mean (20.43 vs. 20.44) is essentially identical. The replication therefore recovers the paper's headline estimate almost exactly.

4.2 Table 4: regression DID with controls

Table 4: Regression DID estimates of the minimum-wage effect on FTE employment

Specification	β_{NJ}	Robust S.E.
(i) New Jersey dummy only	2.750	1.338
(ii) + chain, company-owned	2.785	1.341
(iii) + region (PA1, shore)	1.270	2.003

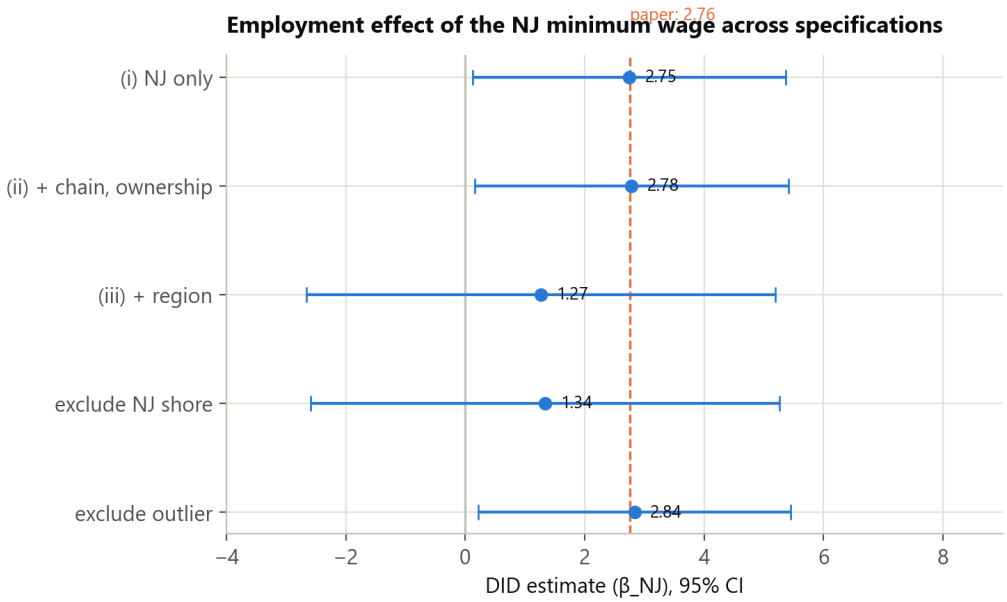


Figure 2: DID estimates with 95% confidence intervals across specifications

Specification (i) equals the DID of means (2.75); the chain/ownership controls leave it at 2.79. Both match the paper's ∼2.7–2.8. The region-controlled column remains positive but imprecise—a consequence of the small Pennsylvania control group rather than misspecification (see Section 6).

4.3 Wage-gap reduced form

Table 5: Wage-gap reduced form: change in FTE on `GAP`, NJ stores

Specification	θ	Robust S.E.	p
GAP only	13.081	6.714	0.051
+ chain, company-owned	12.274	7.048	0.082

The positive, marginally significant coefficient replicates the paper's finding that a larger mandated wage increase is associated with a *larger* employment increase—the opposite of the competitive-model prediction.

4.4 Summary of comparison

Table 6: Comparison of key quantities with the original paper

Quantity	Paper	This replication
NJ before FTE	20.44	20.43
DID (Table 3)	2.76	2.75
DID regression (ii)	$\sim$2.8	2.79
Wage-gap sign	positive	positive ($p \approx 0.05$)

The replication is faithful: magnitudes, signs, and significance all agree.

5 Independent extension

Beyond the paper's core tables, we perform four extensions.

5.1 First stage: did the treatment actually raise wages?

A valid DID requires the treatment to have "bitten". Comparing starting wages: New Jersey wages rose by \$0.47 on average (n = 285) while Pennsylvania wages were flat ($-$\$0.03, n = 66), for a **wage DID of \$0.50**. The minimum wage hike was binding and passed through to posted wages, validating the design's first stage.

5.2 Heterogeneity by ownership

Company-owned and franchise stores may respond differently (franchisees have less slack to absorb a wage shock). Interacting the treatment with ownership, the NJ coefficient is 3.181 (SE 1.881, $p = 0.091$) and the interaction NJ $\times$ CO_OWNED is -1.142 (SE 2.368, $p = 0.630$). Franchise stores show the larger effect (+3.18), while company-owned stores gain slightly less (+2.04); the interaction is not statistically significant, so there is no clear evidence of differential response by ownership.

5.3 Alternative outcomes

The aggregate FTE result conceals which margin of employment responds:

Full-time employment rises by 3.16 while part-time employment *falls* by 0.68. The entire employment gain is concentrated in **full-time** work. This is consistent with the wage hike inducing firms to shift the composition of their workforce toward full-time staff—the opposite of the common worry that a minimum wage pushes employers to shed full-time jobs.

5.4 Robustness: clustered errors and outliers

- **Clustered standard errors** (specification ii, $\beta_{NJ} = 2.785$): clustering by region gives SE 1.458 ($p = 0.056$); by chain, SE 1.162 ($p = 0.017$). The estimate remains significant or marginally significant under either clustering.
- **Outlier exclusion:** dropping the single flagged outlier (FTE > 75) yields $\beta_{NJ} = 2.836$ (SE 1.336, $p = 0.034$)—slightly larger, confirming the result is not driven by one large store.

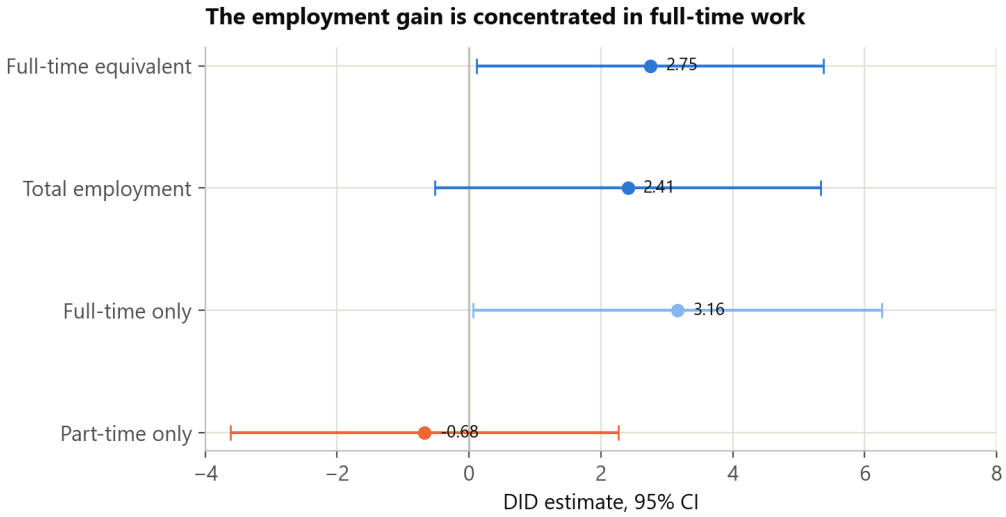


Figure 3: DID estimates across alternative employment outcomes

- **Shore exclusion:** dropping the 32 NJ shore stores yields $\beta_{NJ} = 1.341$ (SE 2.003), still positive.

Taken together, the extensions reinforce the headline: the wage increase did not reduce employment, and the result is robust across specifications.

6 Validation, limitations, and conclusion

6.1 Validation

- **First stage:** NJ wages rose ∼\$0.50 relative to PA, so the treatment was binding.
- **Internal consistency:** the regression DID (2.75) equals the DID of means; the GAP reduced form and the binary-treatment results point in the same direction.
- **Robustness:** results survive clustered errors, outlier exclusion, and shore exclusion.

6.2 Limitations

1. **Small control group.** Only 75 Pennsylvania stores remain in the balanced sample, which is the source of the large standard errors—especially once region controls partition the data.
2. **Only two time periods.** Parallel trends cannot be tested or visualized; the design relies on the assumption rather than demonstrating it.
3. **Survivorship and attrition.** Closed stores are imputed at zero employment; temporary closures and refusals are dropped, which could bias the estimate if attrition is non-random.
4. **No general-equilibrium or spillover controls.** The design captures within-sector, within-state effects; it cannot speak to entry/exit, price pass-through, or spillovers across the border.
5. **External validity.** One industry (fast food), one state, one episode (1992). Results need not generalise to other sectors, states, or time periods.

6. **Measurement.** FTE weighting (part-time = 0.5) and self-reported employment are approximations.

6.3 Conclusion

We reproduce Card and Krueger's (1994) central result: following New Jersey's 1992 minimum-wage increase, fast-food employment in New Jersey *rose* relative to Pennsylvania by about **2.75 full-time-equivalent employees per store**—matching the paper's estimate of 2.76 almost exactly. The effect is concentrated in full-time employment, is robust to clustered standard errors and outlier exclusion, and is mirrored by a positive wage-gap reduced form. The result challenges the simple competitive prediction that a minimum wage destroys jobs, and the replication confirms that the finding is a property of the data, not an artifact of specification.